



Generating Ideas

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If you don't plan for the future, you can't have one

Wrong forecasts of reputable experts (II)

It is not easy to predict the future ... !!

1943	Thomas J. Watson Chief Executive Officer of IBM	„I think there will only be a need on the world market for 5 computers.“
1945	Vannevor Bush Chief Commander of the USA	"I wish, the Americans would finally stop talking about the phantasm intercontinental ballistic missile "
1957	Lee de Forest Inventor of the cathode pipe	„Despite all progress, man will never be able to reach the moon.
1977	Ken Olsen Chief Executive Officer of the computer manufacturer Digital	„I can not see any reason at all, why an individual should have its own computer.“

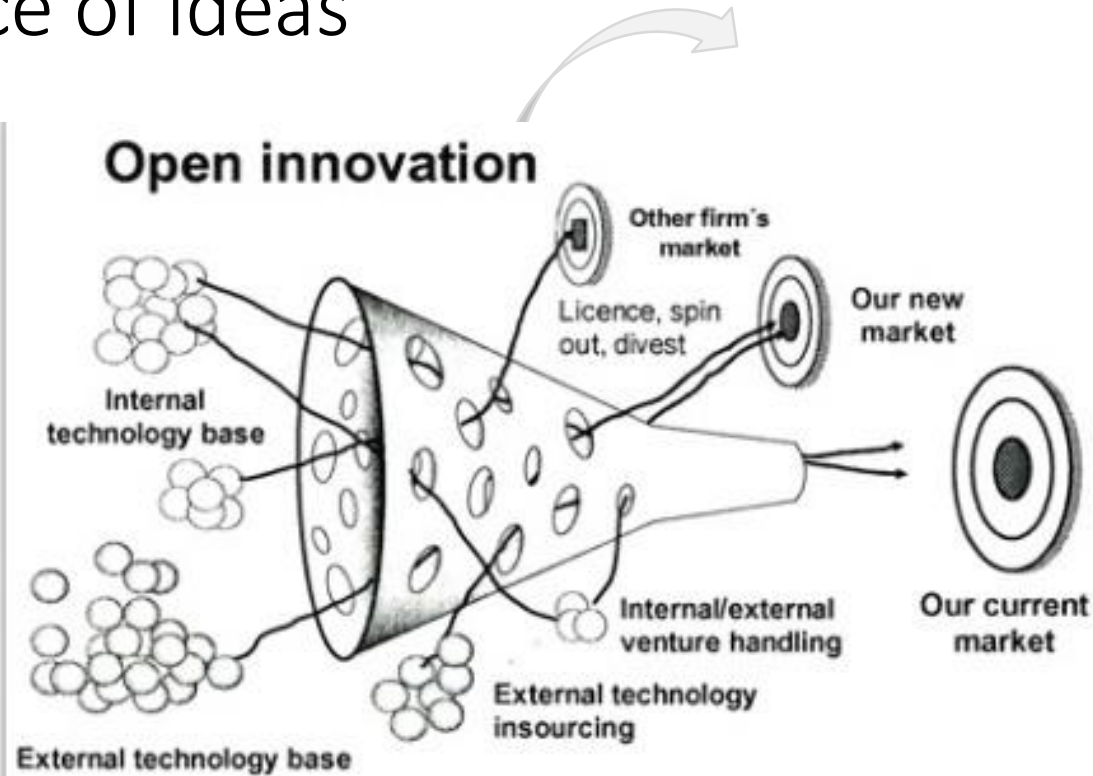
... but you can try to be as sure as possible



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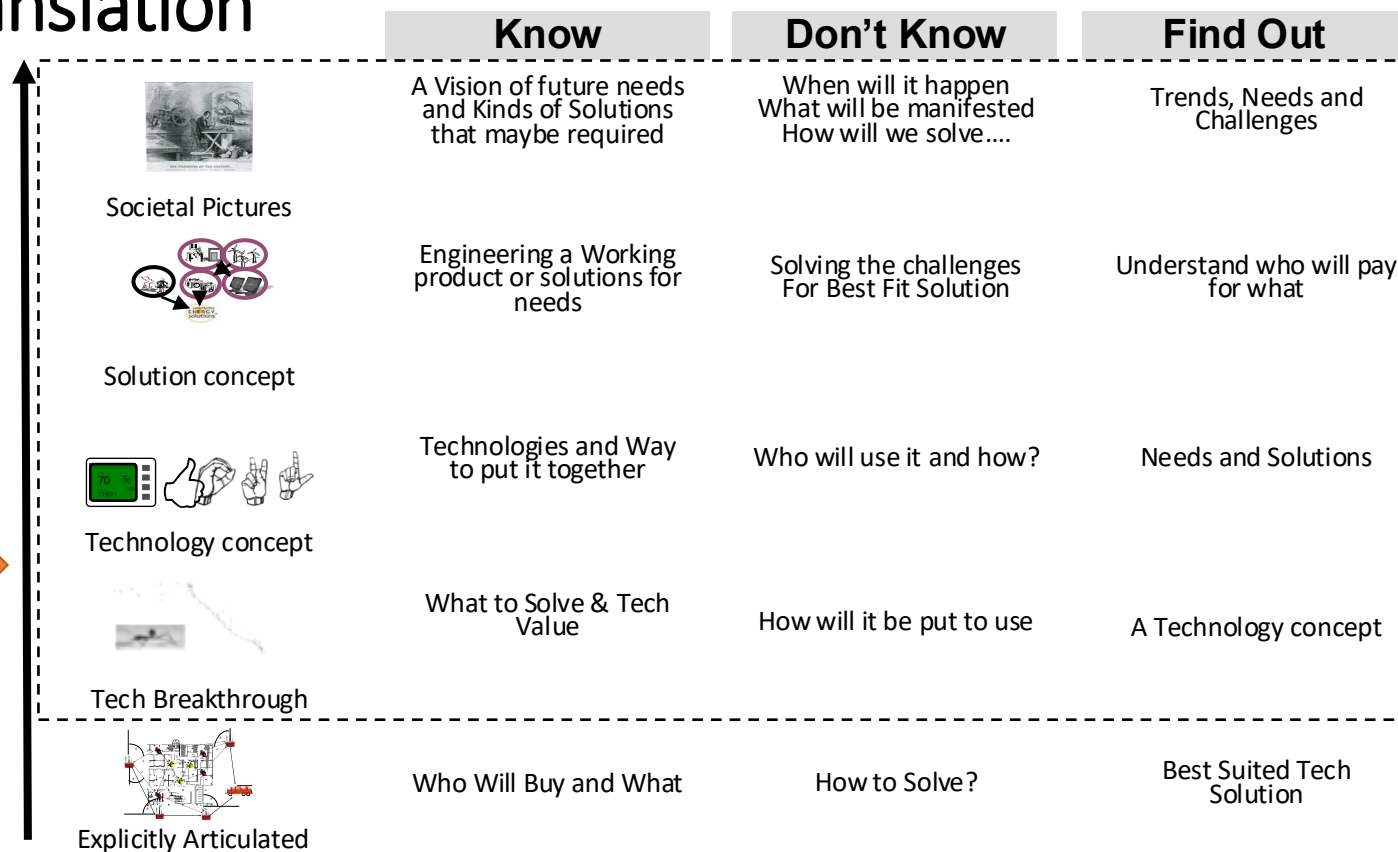
Source of ideas



Stolen with pride from Prof Henry Chesbrough UC Berkeley, Open Innovation: Renewing Growth from Industrial R&D, 10th Annual Innovation Convergence, Minneapolis Sept 27, 2004



A structured approach & frameworks needed to enable the translation



Observe

Connect the dots

Reflective Dialogue

Seed New Ideas



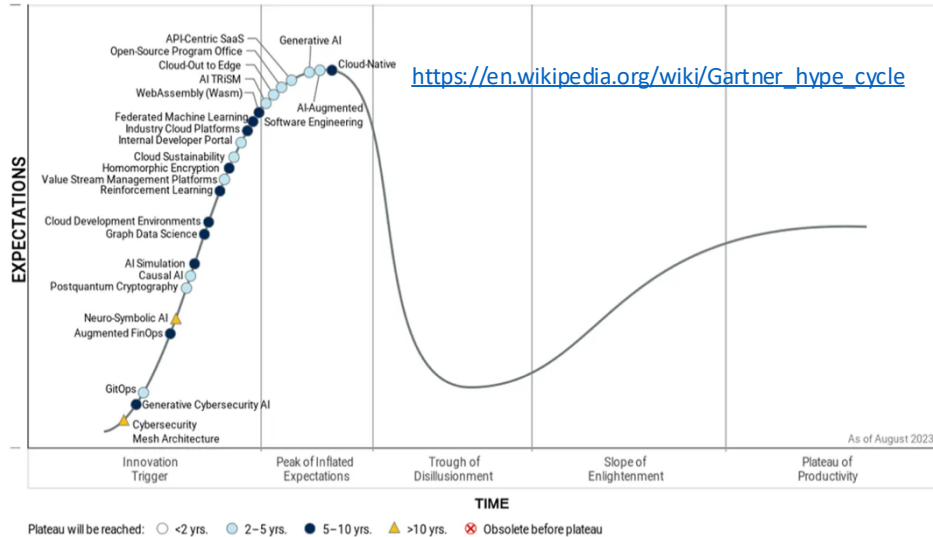
Part 1: The Landscape

Reading the Map of Opportunity



25 must-know emerging technologies designed to help enterprise architecture and technology innovation leaders (2023)

The four main themes



Emergent AI :

Generative AI, Causal AI, AI Simulations, Federated ML, Reinforced Learning, Graph Data Science, Neuro Symbolic AI

Developer experience :

Augmented Software engineering, API Centric SAAS, Internal Developer portals, GitOps, Open Source Program Office

Pervasive cloud, and

Augmented FinOps, Cloud Dev Environments, Cloud Sustainability, Cloud Native, Cloud out to Edge, Web Assembly (wasm)

Human-centric security and privacy

CyberSecurity Mesh Arch, Generative Cybersecurity AI, Homomorphic encryption, Post Quantum Cryptography

Gartner.

- **Evaluate** the business impact of emerging technologies
- **Examine** and explore potentially transformative technologies
- **Strategize** how to benefit from these technologies

<https://www.gartner.com/en/articles/what-s-new-in-the-2023-gartner-hype-cycle-for-emerging-technologies>



Gartner's 3C framework for trend spotting

- Convergence

- ❖ Technologies in Parallel Existence coming together to create exponential value
- ❖ Ex : Mobile, social, cloud and data, and these capabilities coming together to create entirely new types of experiences that were data-rich and connected people together. Cloud created the scale.
- ❖ Happening now is a convergence of, say, mixed reality and IoT, knowledge graphs and AI.
- ❖ Scenario : “Technician on an Airplane”

- Catalysis

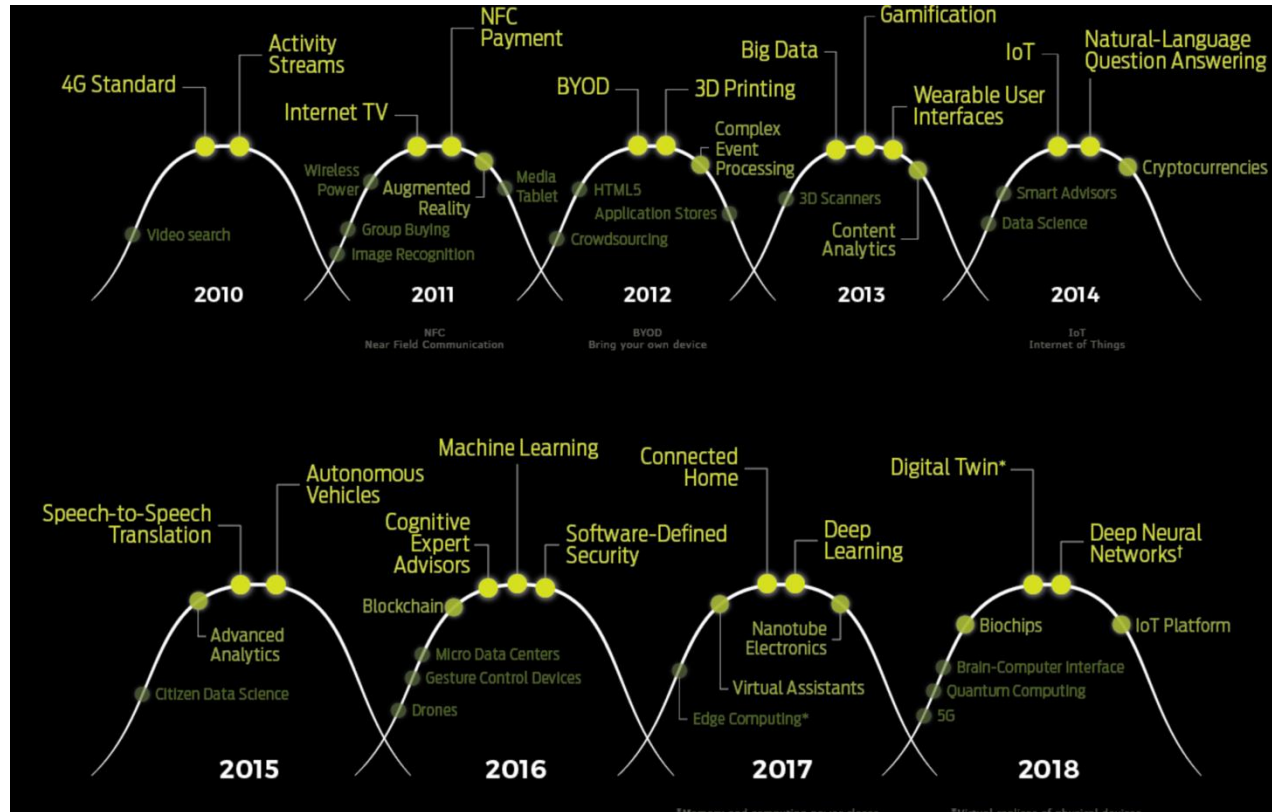
- ❖ Technologies exist in parallel but an event makes it all usable
- ❖ Ex: conversational interfaces, machine learning, neural networks, GPUs for compute. The catalyst : ChatGPT
- ❖ What catalysis role did iPhone play?

- Context

- ❖ Context exists at different levels, trends like political, social-cultural trends, economic, regulatory etc.
- ❖ Ex Impact of conflict in Europe and enterprise strategy. AI regulation in a country and impact on adoption
- ❖ Think of a regulation or a economic trend that kicked off adoption of technology



Hype Cycles Change : Need ears on the ground



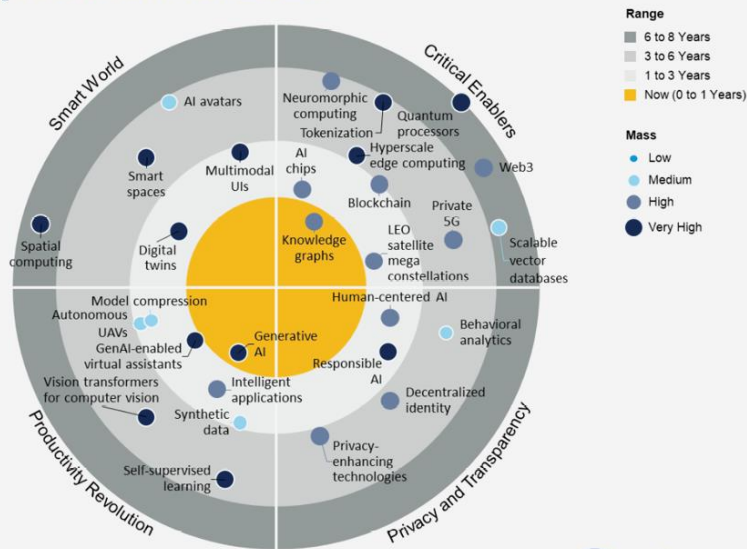
<https://www.visualcapitalist.com/technology-hype-cycles-2000-2018/>



Impact Radar Indicates broad disruptions across markets

The four main themes

Impact Radar for 2024



Source: Gartner
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Gartner

Smart Worlds:

AI Avatars, Multi modal interfaces, digital twins, smart spaces, Spatial computing

Productivity Revolution:

Model compression, autonomous UAVs, GenAI, GenAI enabled Virtual assistants, Vision transformers for CV, Intelligent applications, Synthetic data, Self Supervised learning.

Privacy and Transparency:

Human Centered AI, Behavioural Analytics, Responsible AI, Decentralised Identity, Privacy Enhancing Technologies

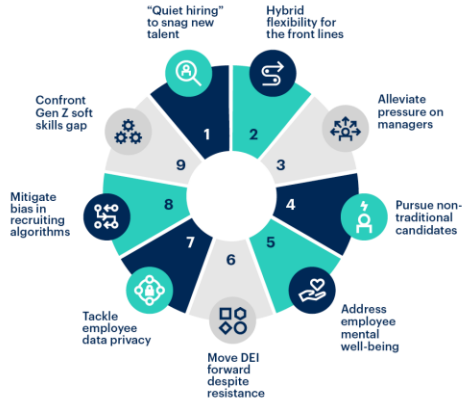
Critical Enablers

Neuromorphic computing, Tokenizers, Quantum processors, Hyperscale edge computing, AI Chips, Web3, block Chain, Knowledge graphs, Private 5G, Scalable vector Databases, LEO Satellite Mega Constellations



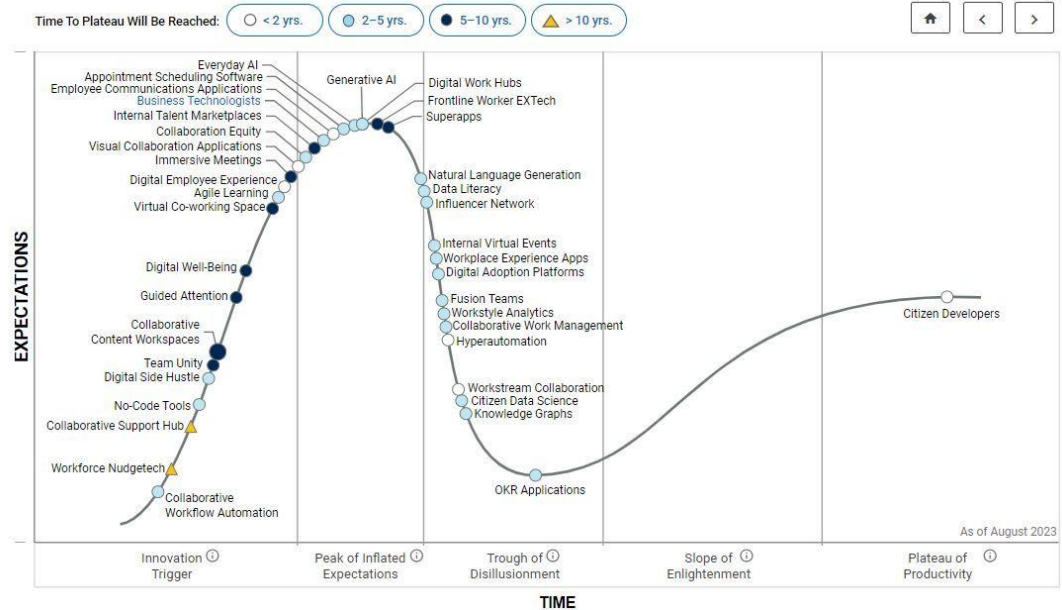
Example-1: Future of workplace and Hype Curve

9 Future of Work Trends in 2023



gartner.com

Gartner.



Gartner : <https://www.gartner.com/en/articles/9-future-of-work-trends-for-2023>

Virgin : <https://www.virgin.com/virgin-unite/latest/five-things-you-need-to-know-about-the-future-of-work>

World economic forum : <https://www.linkedin.com/pulse/future-jobs-report-2023-overview-pravin-shinde-t3ltf/>

Analysts : <https://dionhinchcliffe.com/2023/07/26/the-future-of-work-in-2024-navigating-through-uncertainties/>



Example 2: Map Business Goals To Technology Trends To Develop Short term and Long Term Strategies

Technology Trends Relate to a Range of Enterprise Strategies

Choose your business goal ↓	Sustainable Technology	AI TRISM	Wireless-Value Realization	Industry Cloud Platforms	Digital Immune System	Platform Engineering	Superapps	Adaptive AI	Applied Observability	Metaverse
	Now	0-1 year			1-2 years			2-3 years		
Create secure foundations		●	●		●	●		●		
Maximize value from data		●	●					●	●	
Protect and grow your brand	●	●			●					●
Attract and retain talent	●					●	●			●
Grow revenue	●			●			●			●
Accelerate digital			●	●	●	●			●	





Part 2: The Cauldron

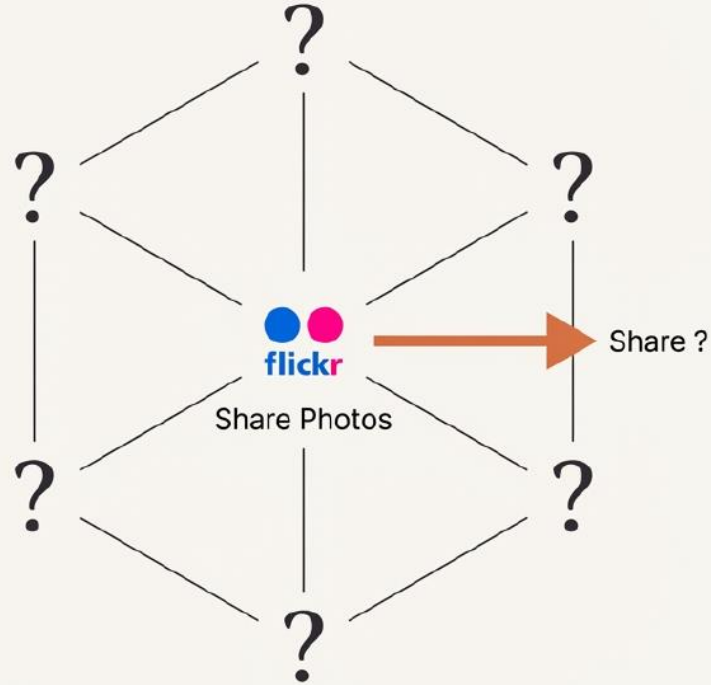
Forging New Concepts from Raw Material



Idea Hexagon: after X, what neXt

Idea Stretching: Six Ways to Evolve Your Core Concept

A single idea is a starting point, not a destination. The Idea Hexagon is a powerful framework for systematically exploring six adjacent possibilities, turning one concept into many. Given idea 'X,' what comes 'neXt'?



The Idea Hexagon: A Practical Guide

1.

1. Generalize (neXt = X^d)

Generalize your concept to another dimension or domain.

Example: Data → Text → Audio → Images → Video.

2.

2. Fusion (neXt = $X+Y$)

Combine your idea with a seemingly dissimilar concept or technology.

Example: Phone + Camera.

3.

3. Find the Nails (neXt = $X\uparrow$)

Given a solution (a hammer), find all the different problems (nails) it can solve.

Example: A new material is invented; what are all its potential applications?

4.

4. Find the Hammers (neXt = $X\downarrow$)

Given a problem (a nail), find all the different ways (hammers) to solve it.

Example: The problem is last-mile delivery; solutions could be drones, robots, gig workers, etc.

5.

5. Add an Adjective (neXt = $X++$)

Modify your core idea with a descriptive quality, often a current trend.

*Example: Banking → **Mobile** Banking, **AI-powered** Banking.*

6.

6. Do the Opposite (neXt = $\bar{X}$)

Invert the core assumption or premise of your idea.

Example: Instead of people coming to a central office, the office comes to the people (remote work).





Part 3: The Crucible

Separating Viable Ideas from Wishful Thinking



The First Reality Check: How Critical is the Need?

Before diving deep, perform this simple “finger in the air” test, developed by HBS Senior Lecturer Janet Kraus, to assess how vital your solution is to potential customers.



Oxygen

A must-have product or service that customers cannot live without. Solves a mission-critical problem.

B2B Example: Salesforce.com.



Aspirin

Eases a significant pain point but is not essential for survival. A “nice-to-have” that makes life or work easier.

Example: Coffee.



Jewelry

A luxury or indulgence. Not vital and doesn't solve a major pain point, but is highly desired for emotional reasons.

Example: Video games.



The Five Filters of a High-Potential Idea

A truly robust idea must stand up to scrutiny from multiple angles. Assess your concept against these five critical dimensions.

Customer Filter

- Do you have a clearly identifiable customer?
- Are you solving a meaningful problem for them?

Product/Service Filter

- Does the idea have a tight niche focus?
- Is a lean methodology viable to build it?

Economic Filter

- Can you achieve healthy margins?
- What are the cash flow requirements?

Timing Filter

- Does it align with a major secular trend?
- Is there a recent innovation enabler?

Competition Filter

- Is the competition limited?
- Does your team have a unique fitness to win?





Part 4: The Compass

Navigating Your Path Forward



Your Innovation Compass: Constantly Monitor the Signposts

A great idea requires constant nurturing and awareness of the changing landscape. Use these questions as your guide to stay ahead.



Technology Study

- Where are we on the technology maturity cycle?



Market Trends

- What are the critical drivers and needs for the technology?



Competitor Trends

- Who are the main industry participants and what are their key areas of focus?



IP Analysis

- What are the grey areas where IP does not yet exist?



External Innovation

- What technology advantage can we leverage by working with universities or open innovation networks?



Startup Ecosystem

- Where are the VC, PE, and Angel investments going?



Your Turn: From Theory to Action

Knowledge is only potential power. Action is where value is created. Apply the frameworks from this guide to your own ideas.

1. Start

Begin with three of your own initial ideas.

2. Expand

Apply the six dimensions of the **Idea Hexagon** to each one, generating up to 18 new variant concepts.

3. Select

From the expanded list, use the **Crucible** filters (Oxygen Test & The Five Filters) to select your top three ideas, and be prepared to justify why they are superior.



Assessment Worksheet

i. Customer Filter	II. Product/Service Filter	iii. Economic Filter
- Clearly Identifiable Customer [A-F] - Meaningful Problem to Solve [] - Segmentable Market [] - Customer Accessibility [] - Customer Loyalty [] Overall Customer Score _____	- Tight Niche Focus [] - No Network Effect [] - Lean Method Viable [] - Team-to-Market Fit [] - Inherent Story (Virality) [] Overall Product Score _____	- Healthy Margins [] - Demand Constraints [] - Supply Constraints [] - Sunk Costs [] - Cash Flow Requirements [] Overall Financing Score _____
IV. Timing Filter	V. Competition Filter	Overall Score
- Secular Trend Alignment [] - Recent Innovation Enabler [] - Market Inefficiency [] - Recent Competition Surge [] - Signs of Commoditization [] Overall Timing Score _____	- Limited Competition [] - Competitor Fitness [] - Team Fitness [] - Defensible Position [] - Barriers to Entry [] Overall Competition Score _____	i. Customer Overall Score [] ii. Product Overall Score [] iii. Financing Overall Score [] iv. Timing Overall Score [] v. Competition Overall Score [] Total Overall Score _____



Thanks



Some Good reads

Innovation Value Chain		Possible Solution
Idea Generation	In-house idea generation	"How to Kill Creativity," by Teresa M. Amabile (HBR September–October 1998) <i>Jamming: The Art and Discipline of Business Creativity</i> , by John Kao (HarperBusiness, 1996)
	Cross-pollination	"Collaboration Rules," by Philip Evans and Bob Wolf (HBR July–August 2005) "Coevolving: At Last, a Way to Make Synergies Work," by Kathleen M. Eisenhardt and D. Charles Galunic (HBR January–February 2000)
	External sourcing	<i>Democratizing Innovation</i> , by Eric von Hippel (MIT Press, 2005) <i>Blue Ocean Strategy</i> , by W. Chan Kim and Renée Mauborgne (Harvard Business School Press, 2004) <i>Open Innovation: The New Imperative for Creating and Profiting from Technology</i> , by Henry Chesbrough (Harvard Business School Press, 2003)
Conversion	Selection	"Bringing Silicon Valley Inside," by Gary Hamel (HBR September–October 1999) <i>Corporate Venturing: Creating New Businesses Within the Firm</i> , by Zenas Block and Ian C. MacMillan (Harvard Business School Press, 1993)
	Development	<i>10 Rules for Strategic Innovators: From Idea to Execution</i> , by Vijay Govindarajan and Chris Trimble (Harvard Business School Press, 2005) <i>The Innovator's Solution: Creating and Sustaining Successful Growth</i> , by Clayton M. Christensen and Michael E. Raynor (Harvard Business School Press, 2003)
Diffusion	Spread of the idea	<i>Payback: Reaping the Rewards of Innovation</i> , by Harold L. Sirkin, James P. Andrew, and John Butman (Harvard Business School Press, 2007) "Tipping Point Leadership," by W. Chan Kim and Renée Mauborgne (HBR April 2003)

